Finding static friction coefficient

1. **Purpose**
   - To determine the statical friction coefficients of a mass moving on inclined plane by the effect of gravity.
2. **Theory**

The sum of all the forces acting on a body is called as the net force. If there is a net force on an object, it is accelerated in the direction of this force. In Figure 1, the motion of a mass m on a frictional inclined plane is shown. Due to the gravitational acceleration (W=mg) and the angle of the slope, this force is shown that it has two components. The motion of the mass on frictional inclined plane is due to the x component of this force.

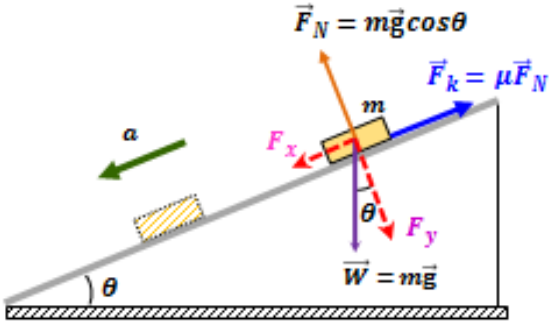


Figure 1. The motion of a mass on inclined plane.

$F_x = mgsin\theta$ - x component

$F_y = mgcos\theta$ - y component

In addition, the y component of the force defines the magnitude of the frictional force owing to the normal force to the mass as formulated below.

$$F_k = \mu F_N$$

The experimental set up is prepared as below.

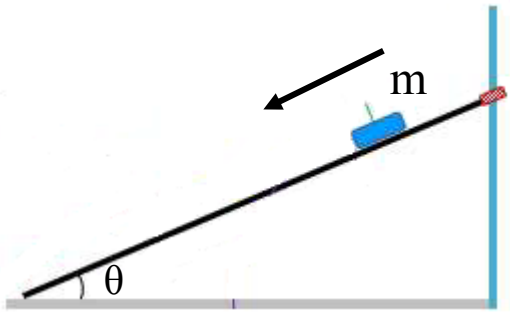


Figure 2. The experimental set-up.

The statical friction coefficient ($\boldsymbol{\mu}$) between the surface of the mass and the surface of the plane will be determined experimentally. To achieve this, the situation of the mass in equilibrium on the plane is examined.

In case of the net force exerted on the mass equals the zero ($\sum \boldsymbol{F_x} = \boldsymbol{0}$), there is a relation between the critical angle ($\boldsymbol{\theta_c}$) where the mass starts to move and statical friction coefficient written below.

$$F_x = F_k$$

$$mgsin\theta = \mu F_N$$

$$mgsin\theta = \mu mgcos\theta$$

$$sin\theta = \mu cos\theta$$

$$\mu = \frac{sin\theta}{cos\theta}$$

$$\boxed{\boldsymbol{\mu = tan\theta}}$$

3. **Experimental Procedure**

- Adjust the board horizontally (θ = 0).
- Place the object at a random position on the board.
- Until the object starts to slide down the board, increase the slope of the inclined plane slowly and record the angle of the board.
- Do this process for three different positions and record the average angle.

$$\boldsymbol{\theta_{avg}} = \frac{\boldsymbol{\theta_1+\theta_2+\theta_3}}{\mathbf{3}}$$

- Calculate the statical friction coefficient according to the average angle.
- Repeat these processes for each surfaces of the object.
- Plot a bar graph of (**μ vs Mass**) and (**μ vs Surface**) for all surfaces.

Table 1. Statical friction coefficients corresponding to the mass and different surfaces.

Surface	**Mass (g)**		**Θ_1 (degree)**	**Θ_2 (degree)**	**Θ_3 (degree)**	**Θ_{avg} (degree)**	**μ**
S_1: Kraft paper	1						
	2						
S_2: Rubber	3						
	4						
S_3: Sand paper	5						
	6						
S_4: Fiberboard	7						
	8						



Figure 3. The bar graph of statical friction coefficients vs masses and surfaces.